

Endogeneity, Instrumental Variables, Two-Stage Least Squares, and GMM

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1. INTRODUCTION AND MOTIVATION

The empirical study of economic relationships rests, almost without exception, on some version of the regression model. Yet the conditions under which regression yields reliable estimates of structural parameters are considerably more stringent than is often acknowledged in applied work. This section lays out the classical framework, identifies the precise point at which it breaks down, and sketches the remedies that the remainder of these notes develop in full.

1.1 The Classical Linear Model and Its Assumptions

Let $\{(y_i, \mathbf{x}_i)\}_{i=1}^n$ be a sample of n observations, where $y_i \in \mathbb{R}$ is the dependent variable and $\mathbf{x}_i \in \mathbb{R}^k$ is a vector of regressors, with the convention that its first component equals unity. The *classical linear model* (CLM) postulates

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta}_0 + \varepsilon_i, \quad i = 1, \dots, n, \quad (1)$$

where $\beta_0 \in \mathbb{R}^k$ is the vector of true parameters and ε_i is an unobservable disturbance. Written in matrix form, (1) becomes $\mathbf{Y} = \mathbf{X}\beta_0 + \varepsilon$, where $\mathbf{Y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times k}$, and $\varepsilon \in \mathbb{R}^n$ collect the observations in the obvious way.

The ordinary least-squares (OLS) estimator is

$$\hat{\beta}_{\text{OLS}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \beta_0 + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon. \quad (2)$$

Its desirable properties depend on the following assumptions, which we state in a form suitable for asymptotic analysis.

Assumption 1.1 (Linearity). The data-generating process satisfies (1) for some $\beta_0 \in \mathbb{R}^k$.

Assumption 1.2 (Random Sampling). $\{(y_i, \mathbf{x}_i)\}_{i=1}^n$ is an independent and identically distributed (i.i.d.) sequence drawn from a well-defined population distribution.

Assumption 1.3 (Full Rank). $\mathbb{E}[\mathbf{x}_i\mathbf{x}_i']$ is finite and positive definite, so that $\text{rank}(\mathbf{X}'\mathbf{X}) = k$ almost surely for n sufficiently large.

Assumption 1.4 (Strict Exogeneity). $\mathbb{E}[\varepsilon_i | \mathbf{x}_i] = 0$ almost surely. As an immediate consequence, $\mathbb{E}[\mathbf{x}_i\varepsilon_i] = \mathbf{0}$ and $\mathbb{E}[\varepsilon_i] = 0$.

Assumption 1.5 (Spherical Disturbances). $\mathbb{E}[\varepsilon_i^2 | \mathbf{x}_i] = \sigma^2 \in (0, \infty)$ and $\mathbb{E}[\varepsilon_i\varepsilon_j | \mathbf{x}_i, \mathbf{x}_j] = 0$ for all $i \neq j$.

Remark. Assumption 1.4 is the pivot around which the entire analysis turns. It is substantially stronger than the *mean independence* condition $\mathbb{E}[\varepsilon_i | \mathbf{x}_i] = 0$; the latter implies $\text{Cov}(\mathbf{x}_i, \varepsilon_i) = \mathbf{0}$, but the converse does not hold in general. The analysis below requires only the zero-correlation form, $\mathbb{E}[\mathbf{x}_i\varepsilon_i] = \mathbf{0}$, for consistency of OLS. We invoke the stronger form only where finite-sample results are concerned.

Under Assumptions 1.1–1.4, the Gauss–Markov theorem establishes that OLS is the best linear unbiased estimator (BLUE) of β_0 . Consistency and asymptotic normality follow from a standard application of the law of large numbers and the central limit theorem.

Theorem 1.6 (Consistency and Asymptotic Normality of OLS). *Under Assumptions 1.1–1.4 and the moment condition $\mathbb{E}[\|\mathbf{x}_i\|^2 \varepsilon_i^2] < \infty$,*

- (i) $\hat{\beta}_{\text{OLS}} \xrightarrow{p} \beta_0$; and
- (ii) $\sqrt{n}(\hat{\beta}_{\text{OLS}} - \beta_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma_{\text{OLS}})$, where $\Sigma_{\text{OLS}} = (\mathbb{E}[\mathbf{x}_i\mathbf{x}_i'])^{-1} \mathbb{E}[\varepsilon_i^2 \mathbf{x}_i\mathbf{x}_i'] (\mathbb{E}[\mathbf{x}_i\mathbf{x}_i'])^{-1}$.

Under the additional Assumption 1.5, Σ_{OLS} simplifies to $\sigma^2(\mathbb{E}[\mathbf{x}_i\mathbf{x}_i^\top])^{-1}$.

Proof. From (2),

$$\hat{\beta}_{\text{OLS}} - \beta_0 = \left(\frac{1}{n} \mathbf{X}'\mathbf{X} \right)^{-1} \frac{1}{n} \mathbf{X}'\boldsymbol{\varepsilon}.$$

By the LLN (Assumption 1.2), $n^{-1}\mathbf{X}'\mathbf{X} \xrightarrow{p} \mathbb{E}[\mathbf{x}_i\mathbf{x}_i^\top]$, which is invertible by Assumption 1.3, and $n^{-1}\mathbf{X}'\boldsymbol{\varepsilon} \xrightarrow{p} \mathbb{E}[\mathbf{x}_i\varepsilon_i] = \mathbf{0}$ by Assumption 1.4. Part (i) follows from Slutsky's theorem. For part (ii), write $n^{-1/2}\mathbf{X}'\boldsymbol{\varepsilon} = n^{-1/2} \sum_{i=1}^n \mathbf{x}_i\varepsilon_i$. The summands are i.i.d. with mean zero and covariance matrix $\mathbb{E}[\varepsilon_i^2\mathbf{x}_i\mathbf{x}_i^\top]$; the CLT yields $n^{-1/2}\mathbf{X}'\boldsymbol{\varepsilon} \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbb{E}[\varepsilon_i^2\mathbf{x}_i\mathbf{x}_i^\top])$. The delta method (or Slutsky) completes the argument. \$■\$

1.2 Endogeneity: Definition, Sources, and Consequences

The entire edifice of Theorem~1.6 collapses the moment Assumption~1.4 is violated.

Definition 1.7 (Endogeneity). A regressor x_{ji} (the j -th component of $\mathbf{x}_i$) is said to be *endogenous* if $\text{Cov}(x_{ji}, \varepsilon_i) \neq 0$, or, more generally, if $\mathbb{E}[\mathbf{x}_i\varepsilon_i] \neq \mathbf{0}$. The model (1) is then said to suffer from *endogeneity*.

Endogeneity is not a single pathology but a family of related problems sharing a common algebraic signature. We distinguish three principal sources.

1.2.1 Omitted Variable Bias

Suppose the true model is

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta}_0 + \gamma_0 c_i + u_i, \quad \mathbb{E}[u_i | \mathbf{x}_i, c_i] = 0,$$

where c_i is an unobserved confounder correlated with $\mathbf{x}_i$. If we estimate the misspecified model (1) with $\varepsilon_i = \gamma_0 c_i + u_i$, then $\mathbb{E}[\mathbf{x}_i\varepsilon_i] = \gamma_0 \mathbb{E}[\mathbf{x}_i c_i] \neq \mathbf{0}$ whenever $\gamma_0 \neq 0$ and c_i is correlated with at least one regressor. This is the *omitted variable bias* (OVB) problem, arguably the most common form of endogeneity in applied economics.

1.2.2 Simultaneity Bias

In many economic settings, y_i and one or more components of $\mathbf{x}_i$ are determined jointly by a system of equations. The canonical example is a supply-and-demand system:

$$\begin{aligned} q_i^d &= \alpha_0 + \alpha_1 p_i + \alpha_2 \mathbf{w}_i' \boldsymbol{\delta} + u_i^d \quad (\text{demand}), \\ q_i^s &= \beta_0 + \beta_1 p_i + \beta_2 \mathbf{r}_i' \boldsymbol{\phi} + u_i^s \quad (\text{supply}), \\ q_i^d &= q_i^s \quad (\text{market clearing}). \end{aligned}$$

In equilibrium, price p_i is a function of both structural errors u_i^d and u_i^s . Regressing quantity on price therefore yields an estimator that is inconsistent for either structural parameter.

1.2.3 Measurement Error

Let x_i^* be the true value of a regressor, observed with error: $x_i = x_i^* + v_i$, where v_i is classical measurement error with $\mathbb{E}[v_i] = 0$, $\text{Cov}(x_i^*, v_i) = 0$. Substituting into the true model $y_i = \beta_0 + \beta_1 x_i^* + u_i$ gives

$$y_i = \beta_0 + \beta_1 x_i + \underbrace{(u_i - \beta_1 v_i)}_{= \varepsilon_i}.$$

Since $\text{Cov}(x_i, \varepsilon_i) = -\beta_1 \text{Var}(v_i) \neq 0$, OLS is inconsistent. The resulting *attenuation bias* pushes the estimate of β_1 toward zero in absolute value.

The consequences of endogeneity are stark.

Theorem 1.8 (Inconsistency of OLS under Endogeneity). *Suppose Assumption 1.4 fails, so that $\boldsymbol{\delta} \equiv \mathbb{E}[\mathbf{x}_i \varepsilon_i] \neq \mathbf{0}$. Then, under Assumptions 1.1–1.3,*

$$\text{plim}_{n \rightarrow \infty} \hat{\boldsymbol{\beta}}_{\text{OLS}} = \boldsymbol{\beta}_0 + (\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top])^{-1} \boldsymbol{\delta} \neq \boldsymbol{\beta}_0.$$

Moreover, no amount of additional data can eliminate this asymptotic bias.

Proof. By the LLN applied to (2),

$$\text{plim } \hat{\boldsymbol{\beta}}_{\text{OLS}} = \boldsymbol{\beta}_0 + \left(\text{plim } \frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \left(\text{plim } \frac{\mathbf{X}'\boldsymbol{\varepsilon}}{n} \right) = \boldsymbol{\beta}_0 + (\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top])^{-1} \boldsymbol{\delta}.$$

Because $\boldsymbol{\delta} \neq \mathbf{0}$ by hypothesis and $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top]$ is non-singular by Assumption 1.3, the second term is non-zero and persists as $n \rightarrow \infty$. \$■\$

Remark. Theorem 1.8 makes clear that endogeneity is fundamentally an identification problem, not a finite-sample one. Collecting more data merely produces a more precise estimate of the wrong quantity. The solution must address the source of the correlation δ , not the sample size.

Example 1.9 (Returns to Schooling). Consider the classic Mincer earnings equation,

$$\log w_i = \beta_0 + \beta_1 S_i + \mathbf{x}_i^\top \gamma + \varepsilon_i,$$

where w_i is the wage, S_i is years of schooling, and $\mathbf{x}_i$ contains observable controls (age, region, etc.). The parameter of interest, β_1 , measures the average causal return to an additional year of education. The problem is that S_i is almost certainly correlated with unobserved *ability* a_i , which also affects wages and is absorbed into ε_i . OLS therefore overstates β_1 . Card [3] addresses this by instrumenting S_i with proximity to a college, a variable that plausibly affects schooling attainment without directly influencing wages — an application of the IV strategy introduced in Section ??.

1.3 A Preview of Solutions

Three broad strategies have been developed to recover consistent estimates of β_0 in the presence of endogeneity. These notes treat the first two in depth and briefly touch on their connections to the third.

Instrumental Variables (IV). The IV approach introduces a vector of *instruments* $\mathbf{z}_i \in \mathbb{R}^\ell$ that satisfy two conditions: *relevance*, $\text{Cov}(\mathbf{z}_i, \mathbf{x}_i) \neq \mathbf{0}$, and *exogeneity*, $\mathbb{E}[\mathbf{z}_i \varepsilon_i] = \mathbf{0}$. These two conditions together restore identification. Section ?? develops the IV estimator, establishes its asymptotic properties, and discusses the critical practical issue of *instrument strength*.

Two-Stage Least Squares (2SLS). When $\ell > k$ — i.e. when there are more instruments than endogenous regressors — the model is *overidentified* and IV is no longer unique. 2SLS provides an optimal way to exploit all available instruments simultaneously. Section ?? derives the 2SLS estimator, relates it to IV in the just-identified case, and establishes its efficiency within the class of IV estimators under homoskedastic errors.

Generalized Method of Moments (GMM). GMM nests both IV and 2SLS as special cases. It begins from a set of moment conditions $\mathbb{E}[\mathbf{g}(\mathbf{w}_i, \beta_0)] = \mathbf{0}$ and constructs the estimator by minimising a quadratic form in the sample analogue. With an *optimal* weighting matrix, GMM achieves the semiparametric efficiency bound within the class of estimators exploiting the same moment conditions. Crucially, GMM extends to heteroskedastic and autocorrelated errors without altering the estimating equations — a feature that makes it indispensable in applied work. Section ?? provides a self-contained treatment.

Diagnostics. Identifying endogeneity in a given dataset, assessing instrument validity, and testing overidentifying restrictions are tasks of independent importance. Section ?? covers the Hausman test, the Sargan–Hansen J -statistic, and the Stock–Yogo weak-instrument test.

Exercise 1.1. Consider the bivariate model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with $\text{Cov}(x_i, \varepsilon_i) = \sigma_{x\varepsilon} \neq 0$ and $\text{Var}(x_i) = \sigma_x^2$.

- (i) Show that $\text{plim } \hat{\beta}_{1,OLS} = \beta_1 + \sigma_{x\varepsilon}/\sigma_x^2$.
- (ii) Determine the sign of the asymptotic bias when (a) ability is the omitted variable in the schooling example, and (b) classical measurement error is present.
- (iii) Suppose you observe a variable z_i with $\text{Cov}(z_i, x_i) = \sigma_{zx} \neq 0$ and $\text{Cov}(z_i, \varepsilon_i) = 0$. Propose an estimator for β_1 based on these covariances, and verify that it is consistent.

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